Total No. of Questions : 8]

SEAT No. :

PE4276

[Total No. of Pages : 2

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S.E. (Artificial Intelligence and Data Science)

OPERATING SYSTEMS

(2019 Pattern) (Semester-III) (217521)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q6, Q.7 or Q.8.***
2) ***Figures to the right indicates full marks.***
3) ***Assume suitable data if necessary.***

Q1) a) What is deadlock? Explain necessary conditions for deadlock. **[6]**

b) Give a semaphore solution for producer-consumer problem. **[6]**

c) Short note on Mutual exclusion. **[5]**

OR

Q2) a) Write a semaphore solution for readers-writers problem. **[6]**

b) Explain different inter-process communication mechanism. **[6]**

c) What is deadlock? Write deadlock prevention policies. **[5]**

Q3) a) Explain: **[6]**

i) Internal fragmentation

ii) External fragmentation

b) Explain dynamic partitioning placement algorithms. **[6]**

c) Write a short note on Buddy System. **[5]**

OR

Q4) a) Explain fixed and dynamic memory partitioning. **[6]**

b) Explain the concept of Virtual memory. **[6]**

c) Explain paging mechanism. **[5]**

P.T.O.

Q5) a) Explain different file organization techniques. **[6]**

b) Explain free space management techniques. **[6]**

c) List and Explain directory management techniques. **[6]**

OR

Q6) a) Write a short note on Directory Structure. **[6]**

b) List and Explain any two file allocation mechanisms. **[6]**

c) Write a short note on I/O buffering. **[6]**

Q7) a) How Memory management is done in Linux. **[6]**

b) Write short note on Linux file system. **[6]**

c) Explain Process management in Linux. **[6]**

OR

Q8) a) Explain Process Scheduling in Linux. **[6]**

b) State and explain different Linux inter-process communication mechanism. **[6]**

c) Write short note on Kernel Modules. **[6]**

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